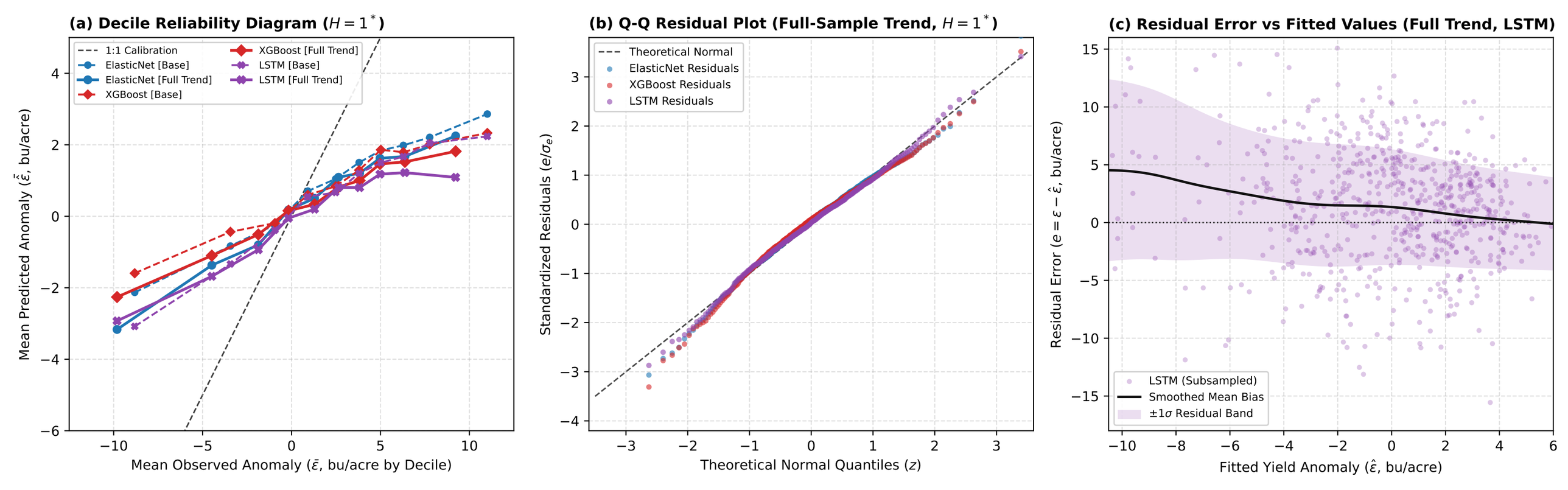

(a) Decile Reliability Diagram ($H = 1^*$)
1:1 Calibration
ElasticNet [Base]
ElasticNet [Full Trend]
XGBoost [Base]
XGBoost [Full Trend]
LSTM [Base]
LSTM [Full Trend]
Mean Predicted Anomaly ($\bar{\hat{\varepsilon}}$, bu/acre)
Mean Observed Anomaly ($\bar{\varepsilon}$, bu/acre by Decile)
(b) Q-Q Residual Plot (Full-Sample Trend, $H = 1^*$)
Theoretical Normal
ElasticNet Residuals
XGBoost Residuals
LSTM Residuals
Standardized Residuals (e/σ_e)
Theoretical Normal Quantiles (z)
(c) Residual Error vs Fitted Values (Full Trend, LSTM)
LSTM (Subsampled)
Smoothed Mean Bias
±1σ Residual Band
Residual Error ($e = \varepsilon - \hat{\varepsilon}$, bu/acre)
Fitted Yield Anomaly ($\hat{\varepsilon}$, bu/acre)